

$$\begin{aligned}
& \mathbf{C}_\phi 2^2 \rightarrow -\hat{\mathbf{m}}_{\mathbf{H}_d} + \\
& \frac{1}{16\pi^2} \left(-\frac{1}{2} \sum_{\mathbf{p}} g_1^2 \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{d}}^{\mathbf{p}}] + 3 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{d}}^{\mathbf{r}}] - \frac{1}{2} \sum_{\mathbf{p}} g_1^2 \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{e}}^{\mathbf{p}}] + \overline{y_e^{\mathbf{p}r}} y_e^{\mathbf{p}r} \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{e}}^{\mathbf{r}}] + \right. \\
& \left(\overline{y_e^{\mathbf{p}r}} y_e^{\mathbf{p}r} + \frac{1}{2} \sum_{\mathbf{p}} g_1^2 \right) \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{l}}^{\mathbf{p}}] + \left(3 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} - \frac{1}{2} \sum_{\mathbf{p}} g_1^2 \right) \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \\
& \sum_{\mathbf{p}} g_1^2 \mathbf{L}F_{1,0}[\mathbf{m}_{\tilde{u}}^{\mathbf{p}}] - g_1^2 \mathbf{L}F_{1,1,-1}[\mathbf{m}_1, \tilde{\mu}] - 3 g_2^2 \mathbf{L}F_{1,1,-1}[\mathbf{m}_2, \tilde{\mu}] + \\
& \left. 3 \overline{a_d^{\mathbf{p}r}} a_d^{\mathbf{p}r} \mathbf{L}F_{1,1,0}[\mathbf{m}_{\tilde{d}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \overline{a_e^{\mathbf{p}r}} a_e^{\mathbf{p}r} \mathbf{L}F_{1,1,0}[\mathbf{m}_{\tilde{e}}^{\mathbf{r}}, \mathbf{m}_{\tilde{l}}^{\mathbf{p}}] + 3 \tilde{\mu}^2 \overline{y_u^{\mathbf{p}r}} y_u^{\mathbf{p}r} \mathbf{L}F_{1,1,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] \right)
\end{aligned}$$